

# Generating Fiducial Labels for MedalCare-XL with ECGdeli

*Methodology, Quality Control, and Integration into the Delineation Pipeline*

## 1. Purpose and context

MedalCare-XL is the intended training corpus for the ECG-delineation model. The dataset is distributed as signals only with the 12-lead waveforms and the electrophysiological parameter files used to generate them, but no fiducial (P/QRS/T onset-peak-offset) annotations.

In the interim, we generate fiducial labels with ECGdeli, the open-source delineation toolbox from the same group (KIT-IBT) that produced MedalCare-XL, and the same tool the dataset authors used to extract interval features for their own technical validation (Table 6 of the published paper’s data descriptor).

## 2. The data and the manifest

The official MedalCare-XL dataset release is described as containing approximately 16,900 ECGs. The version processed in this report yielded 16,848 ten-second 12-lead ECGs at 500 Hz, across a healthy class and seven pathologies (myocardial infarction is the bulk, with six occlusion/transmurality sub-classes), with valid, uniquely resolved signal and metadata paths. Each ECG is stored leads-by-samples (12 rows  $\times$  5000 columns, no header) in three versions: raw (noise-free), filtered, and noise-added.

| Disease class  | ECGs  | Note                                                    |
|----------------|-------|---------------------------------------------------------|
| <b>sinus</b>   | 1,300 | healthy control                                         |
| <b>mi</b>      | 7,806 | 6 MI sub-classes (LAD/LCX/RCA $\times$ transmuralities) |
| <b>avblock</b> | 1,300 | 1st-degree AV block                                     |
| <b>lbbb</b>    | 1,300 | left bundle branch block                                |
| <b>rbbb</b>    | 1,298 | right bundle branch block                               |
| <b>lae</b>     | 1,300 | left atrial enlargement                                 |
| <b>fam</b>     | 1,300 | fibrotic atrial cardiomyopathy                          |
| <b>iab</b>     | 1,244 | interatrial conduction block                            |

### 2.1 Manifest (preserving the class information)

To avoid losing the morphology/disease information when the signals are flattened, we first built a single index **medalcare\_manifest.csv**, with one row per ECG, containing `record_id`, `disease_class`, `mi_subclass`, `split`, `batch`, `run_id`,

sampling rate, and the paths to all three signal versions and the parameter files. All 16,848 records resolve to unique IDs and verified paths. This manifest drives every downstream step, so the 84,000 files never have to be handled by hand.

### 3. Delineation with ECGdeli

ECGdeli operates on an in-memory, matrix-oriented samples  $\times$  leads format, whereas MedalCare stores leads  $\times$  samples. Rather than duplicate 27 GB, the driver transposes each file on load. The processing per record is -> read CSV  $\rightarrow$  transpose to  $5000 \times 12 \rightarrow$  (optional isoline correction)  $\rightarrow$  Annotate\_ECG\_Multi (signal, 500, 'PQRST')  $\rightarrow$  map the fiducial-point table (FPT) columns into the project schema  $\rightarrow$  append one row per beat-per-lead. The driver (**run\_ecgdeli\_medalcare.m**) is resumable and was validated on a 20-record smoke test before the full run.

#### 3.1 Fiducial-point table (FPT) column mapping

ECGdeli returns P/QRS/T onsets, peaks, and offsets, where the individual Q, R, and S peaks are measured natively (so Q and S are measured, not derived). Column 6 = R-peak is confirmed in **Annotate\_ECG\_Multi.m** while columns 1/3/4/12 are confirmed in the MedalCare synthesis code.

| FPT col | Fiducial  | FPT col | Fiducial               |
|---------|-----------|---------|------------------------|
| 1       | P onset   | 7       | S peak                 |
| 2       | P peak    | 8       | QRS offset             |
| 3       | P offset  | 10      | T onset                |
| 4       | QRS onset | 11      | T peak                 |
| 5       | Q peak    | 12      | T offset               |
| 6       | R peak    | —       | (U wave: not detected) |

Output Convention - sample indices are 0-based; milliseconds = sample  $\times$  1000 / 500. Disease class and, for infarction, the MI subclass are carried on every row: mi\_subclass holds the detailed label (LAD/LCX/RCA at each severity, with the LCX anterior/posterior variant) and mi\_group holds the normalised six-category group, so morphology information is never lost.

**How Q and S are measured** - “measured” here means the driver reads the Q, R and S positions straight from ECGdeli’s fiducial-point table (columns 5, 6 and 7) as produced by the toolbox’s own QRS-delineation routine so for each beat it simply takes fpt(b,5), fpt(b,6) and fpt(b,7) with no computation of its own.

The Q, R and S positions are taken directly from ECGdeli's QRS-delineation output rather than being derived by the present pipeline. Their accuracy nevertheless remains dependent on ECGdeli and may be reduced in low-amplitude, fractionated or atypical QRS morphologies.

### 3.2 Smoke test and full run

The 20-record smoke test (avblock) produced a physiologically and clinically consistent result, confirming the input orientation, schema mapping, unit conversion, output generation and physiological plausibility before scaling.

| Biomarker (lead II median) | Value  | Interpretation                                                      |
|----------------------------|--------|---------------------------------------------------------------------|
| PQ/clinical PR interval    | 214 ms | prolonged — consistent with 1st-degree AV block (the avblock class) |
| QRS duration               | 132 ms | matches MedalCare's simulated QRS width (paper Table 6)             |
| QT interval                | 386 ms | within range                                                        |
| Heart rate                 | 77 bpm | physiological                                                       |

Schema, ms = sample $\times$ 2, and presence flags all verified with zero errors. The full run then labelled all 16,848 records, producing 2,646,204 beat-lead fiducial rows in **medalcare\_fiducials\_ecgdeli.csv**.

**From 16,848 records to 2,646,204 rows** - the count grows because the schema stores one row per beat per lead, not one per recording. Each ten-second ECG contains on the order of 8-14 heartbeats, and ECGdeli delineates all twelve leads independently, so a single record expands to roughly (beats  $\times$  12) rows. Across the dataset this is about 16,848 records  $\times$  12 leads  $\times$   $\sim$ 13 beats  $\approx$  2.6 million, the exact figure of 2,646,204 is the sum of the actual beats detected in every lead of every record, an average of about 13 beats per lead.

## 4. Quality control of the pseudo-labels

### 4.1 QC design (qc\_review\_list.py)

A streaming Python screen (it processes the 2.65 M-row file without loading it into memory) classifies every beat into three tiers. A small tolerance ignores trivial boundary wobbles, so the review list stays targeted.

**What informs the design, and what an inversion is** - The automated QC screen targets two readily testable classes of annotation failure: violations of expected temporal ordering and derived intervals outside predefined

plausibility ranges. It does not detect every possible localization error, particularly consistently shifted but physiologically plausible boundaries.

**An inversion here means** a pair of adjacent fiducials that appear out of order, meaning a fiducial that should come later is assigned an earlier sample index than the one before it, for example a P offset placed before the P peak, or a T onset placed after the T peak. It is determined by walking the eleven fiducials of each beat in their canonical order (P onset, P peak, P offset, QRS onset, Q, R, S, QRS offset, T onset, T peak, T offset) and comparing every neighbouring pair; whenever the earlier point's sample index exceeds the later point's, that pair is flagged and the size of the gap, in samples, is recorded. That size sets the tier, an inversion of at most three samples ( $\leq 6$  ms) is negligible boundary wobble and left clean, whereas a structural inversion inside the QRS block, a gross inversion beyond twenty samples ( $\approx 40$  ms), or an out-of-range interval is treated as critical.

- **CRITICAL - review or exclude - a structural QRS-order violation, a QRS/QT/PR interval outside a predefined plausibility range, a gross P/T boundary inversion ( $> 20$  samples  $\approx 40$  ms), or a missing R peak.**
- **MINOR - a moderate P/T boundary inversion (3-20 samples), treated as lower priority and left out of the review list.**
- **CLEAN - no tested issue, or an inversion of  $\leq 3$  samples ( $\leq 6$  ms, negligible).**

Two outputs are written -> medalcare\_qc\_review\_list.csv (one row per CRITICAL beat, with its failure reasons and key fiducial samples) and medalcare\_qc\_record\_summary.csv (every record ranked worst-first by critical rate). The full code is in the repository (ecgdeli\_labelling/scripts/qc\_review\_list.py; see the Code and data availability appendix).

## 4.2 QC results

| Tier                            | Beats     | Share   |
|---------------------------------|-----------|---------|
| <b>CRITICAL (review)</b>        | 104,171   | 3.94 %  |
| <b>MINOR (small P/T wobble)</b> | 477,300   | 18.04 % |
| <b>CLEAN</b>                    | 2,064,733 | 78.03 % |

Structural QRS-ordering violations were rare, occurring in 50 beat-lead rows. This indicates that the returned QRS fiducials are generally internally ordered but does not by itself establish their localization accuracy. The critical flags are dominated instead by interval and T-boundary problems - QT out-of-range (52,202), gross P/T boundary inversion (48,977), PQ/clinical PR out-of-range

(40,416) and QRS duration (11,350). These reason counts are not mutually exclusive, because one beat can contribute to more than one category.

Per beat vs per record-lead - The rates above count individual beats, which overstates how much of the dataset needs work. Each MedalCare-XL ECG is synthesised from common underlying P-wave and QRST templates, so the beats of a (record, lead) share the same source morphology but are not exact duplicates: RR timing varies and the QRS-offset-T-offset segment is stretched or compressed. A lone flagged beat among clean siblings may therefore represent a transient delineation glitch rather than a systematically broken record. Counting at the (record, lead) level, 2.10% of units are persistently flagged under the operational threshold of  $\geq 50\%$  of the unit's beats flagged. Approximately 54% of the flagged beats sit in units below this threshold — these are lower-priority for manual review and may be suppressed by an aligned per-record consensus based on intervals or R-aligned relative positions. The recurring patterns collapse to 4,251 persistently flagged units, concentrated in MI/LBBB/RBBB, across 2,300 recordings. Manual review can therefore be targeted to these units rather than all 104,171 flagged beat-lead rows. One representative beat can guide the correction of a unit, but corrected boundaries must be transferred to the remaining beats using R-peak alignment and, where necessary, temporal scaling. The per-beat 3.94% is thus an upper bound on individually flagged rows, while the persistently flagged units provide a more realistic review set; the 50% threshold does not prove that all remaining units are error-free.

| <b>Disease class</b> | <b>Persistent-flag rate (record-lead)</b> | <b>Persistent / total units</b> |
|----------------------|-------------------------------------------|---------------------------------|
| <b>lbbb</b>          | 4.52%                                     | 705 / 15,600                    |
| <b>mi</b>            | 2.73%                                     | 2,558 / 93,672                  |
| <b>rbbb</b>          | 2.04%                                     | 317 / 15,576                    |
| <b>iab</b>           | 1.51%                                     | 225 / 14,928                    |
| <b>fam</b>           | 0.97%                                     | 152 / 15,600                    |
| <b>sinus</b>         | 0.96%                                     | 149 / 15,600                    |
| <b>lae</b>           | 0.84%                                     | 131 / 15,600                    |
| <b>avblock</b>       | 0.09%                                     | 14 / 15,600                     |

### 4.3 Distribution of QC flags by disease class

The per-class critical-flag rate follows the expected pattern whereby the ventricular-pathology classes (LBBB, MI, RBBB) have the highest rates, while the healthy and atrial-disease classes have the lowest. This gives empirical, quantified evidence of where the pseudo-labels need the most scrutiny and the

strongest case for pursuing the simulation-derived ground truth for those classes, but the QC flag rate does not by itself measure true annotation error.

| Disease class  | Critical rate | Critical / total beats |
|----------------|---------------|------------------------|
| <b>lbbb</b>    | 7.8 %         | 16,217 / 208,428       |
| <b>mi</b>      | 4.3 %         | 51,726 / 1,204,620     |
| <b>rbbb</b>    | 4.2 %         | 8,276 / 198,816        |
| <b>iab</b>     | 3.2 %         | 6,325 / 199,392        |
| <b>sinus</b>   | 3.1 %         | 6,590 / 210,480        |
| <b>lae</b>     | 2.9 %         | 6,032 / 209,076        |
| <b>fam</b>     | 2.7 %         | 5,768 / 210,072        |
| <b>avblock</b> | 1.6 %         | 3,237 / 205,320        |

#### 4.4 Population-level validation against the paper’s Table 6

We extracted the same six timing biomarkers from our ECGdeli labels on the healthy sinus cohort, across roughly 200,000 beat-lead observations over all twelve leads, and compared the per-lead means against the simulated values in Table 6 of the published paper. Given that that reference was also produced with ECGdeli, the consistency check here assesses whether we drive the tool similarly to the authors, as we don’t have an independent ground-truth validation. Table 6 is the only per-lead numeric reference the descriptor provides: the preprint gives only the lead-II Figure 5 and the per-class Figure 6, with no equivalent table, so the per-lead comparison below is necessarily against the published paper’s Table 6. As Section 4.6 shows, QT differs systematically between Table 6 and the preprint, so the QT row documents a reference discrepancy rather than independently validating either value.

| Lead | P dur     | QRS dur   | T dur     | PQ int    | QT int           | RR int    |
|------|-----------|-----------|-----------|-----------|------------------|-----------|
| I    | 117 / 124 | 131 / 131 | 171 / 178 | 119 / 128 | <b>379 / 311</b> | 759 / 758 |
| II   | 125 / 128 | 126 / 126 | 182 / 182 | 124 / 127 | <b>386 / 317</b> | 759 / 758 |
| III  | 156 / 165 | 126 / 127 | 182 / 183 | 160 / 172 | <b>379 / 307</b> | 759 / 758 |
| aVR  | 118 / 127 | 128 / 129 | 178 / 179 | 116 / 126 | <b>387 / 319</b> | 759 / 758 |
| aVL  | 138 / 154 | 128 / 129 | 180 / 183 | 145 / 169 | <b>373 / 299</b> | 759 / 758 |
| aVF  | 136 / 141 | 125 / 125 | 183 / 184 | 137 / 143 | <b>381 / 310</b> | 759 / 758 |
| V1   | 136 / 141 | 129 / 129 | 177 / 181 | 154 / 161 | <b>374 / 304</b> | 759 / 758 |
| V2   | 153 / 156 | 136 / 136 | 175 / 176 | 178 / 182 | <b>360 / 288</b> | 759 / 758 |
| V3   | 150 / 154 | 132 / 133 | 179 / 179 | 170 / 174 | <b>359 / 286</b> | 759 / 758 |
| V4   | 137 / 141 | 127 / 127 | 179 / 180 | 144 / 149 | <b>363 / 290</b> | 759 / 758 |
| V5   | 125 / 129 | 124 / 124 | 176 / 177 | 122 / 127 | <b>380 / 311</b> | 759 / 758 |
| V6   | 120 / 123 | 126 / 127 | 174 / 175 | 115 / 119 | <b>387 / 321</b> | 759 / 758 |

Each cell shows our mean vs the published Table 6 “sim” mean, in milliseconds (QT highlighted). The QT reference column is the published Table 6 value; the preprint provides no per-lead QT table, but its Figure 5 (lead II, ~385 ms) is closer to our QT.



The agreement is close for five of the six biomarkers. QRS duration is essentially identical to the published values (mean absolute difference 0.3 ms across all twelve leads), and RR interval (0.8 ms), T duration (1.8 ms), PQ interval (7.5 ms) and P duration (6.1 ms) all reproduce the reference within single-digit-to-low-tens of milliseconds with the larger P-wave gaps falling, as expected, in the low-amplitude P leads (III, aVL). This supports reproducibility of our ECGdeli processing and the dataset's published population statistics, but not absolute fiducial accuracy.

The sixth biomarker, QT, reads +70.6 ms against the published Table 6 (a uniform 66-74 ms across leads). Our QT — about 386 ms in lead II — is close to the value shown in the descriptor's own preprint (Figure 5, ~385 ms), whereas the published Table 6 reports ~317 ms. The available evidence therefore indicates a systematic difference in processing, aggregation, software version, or T-offset convention between the references. Agreement in PQ/clinical PR and QRS duration supports reproducibility of those interval calculations but does not prove that QRS onset or T offset is exact. The QT discrepancy is retained as an unresolved reference difference rather than attributed definitively to either Table 6 or our labels.

#### 4.5 Full reproduction of the population statistics

To turn the Table 6 check into a thorough reproducibility test, we recomputed every population statistic the descriptor reports: the healthy timing means and their standard deviations, the five amplitude features, and the per-class distributions of the descriptor's Figure 6. Timing statistics were computed from the complete set of 2,646,204 beat-lead annotations and then aggregated to one median value per recording and lead. The amplitude analyses used targeted samples, because retrieving voltages requires loading the corresponding raw signal files: 200 healthy recordings for the all-lead comparison and 100 recordings per class for the Figure 6 contrasts. Amplitudes were read as the raw signal voltage at each detected peak relative to a per-beat isoelectric baseline (median of the twelve samples ending at QRS onset). Each MedalCare-XL ECG is synthesised from common P-wave and QRST templates, while successive beats vary in RR timing and in the stretch or compression of the QRS-offset-T-offset segment; aggregating to one value per record matches the paper's per-ECG extraction. The computation was done in Python (pandas/NumPy); an equivalent MATLAB script, `reproduce_paper_stats.m`, is kept alongside the pipeline.

Timing. Five of the six healthy timing biomarkers reproduce the published simulated means to within single-digit-to-low-tens of milliseconds, QRS duration to 0.3 ms. The sixth, QT, reads +70.6 ms against the published Table 6 but is close to the descriptor's preprint (~385 ms; Section 4.6). This unresolved discrepancy should be interpreted as a difference between the references rather than proof that either value is correct. Under the per-record aggregation, the standard deviations also match the paper reasonably closely: QRS duration is identical at 14.7 ms, while QT interval (23.8 ms), T-wave duration (27.7 ms), and RR interval (59.7 ms) remain close to the published values.

| Feature             | mean $ \Delta\mu $ (ms) | mean $\Delta\mu$ (ms) | our $\sigma$ (ms) | paper $\sigma$ (ms) |
|---------------------|-------------------------|-----------------------|-------------------|---------------------|
| <b>P duration</b>   | 6.1                     | -6.1                  | 24.0              | 21.2                |
| <b>QRS duration</b> | 0.3                     | -0.3                  | 14.7              | 14.7                |
| <b>T duration</b>   | 1.8                     | -1.8                  | 27.7              | 26.6                |
| <b>PQ interval</b>  | 7.5                     | -7.5                  | 34.3              | 30.1                |
| <b>QT interval</b>  | 70.6                    | +70.6                 | 23.8              | 24.5                |
| <b>RR interval</b>  | 0.8                     | +0.8                  | 59.7              | 54.4                |

Amplitude. The five amplitude features reproduce the published values to within 0.19 mV (the maximum, on the R wave; all-lead mean absolute differences over a 200-record sample) and recover the expected lead-wise polarity (R positive in I/II/V5/V6, negative in aVR/V1-V3). Residual differences, largest for R, may reflect the baseline definition, lead-specific peak placement, or fiducial differences.

| Feature            | mean $ \Delta $ (mV) | lead sign agreement |
|--------------------|----------------------|---------------------|
| <b>P amplitude</b> | 0.02                 | 12/12               |
| <b>Q amplitude</b> | 0.05                 | 11/12               |
| <b>R amplitude</b> | 0.19                 | 11/12               |
| <b>S amplitude</b> | 0.03                 | 12/12               |
| <b>T amplitude</b> | 0.06                 | 12/12               |

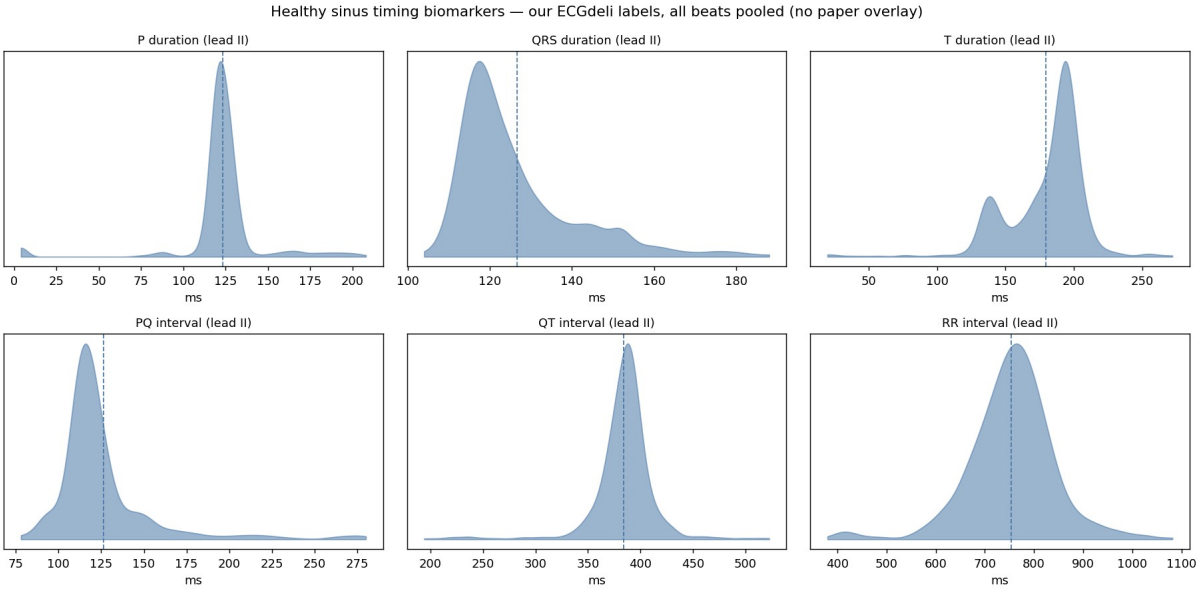
Per-class (Figure 6). Every disease class shifts in the physiologically expected direction: AV block prolongs PR by ~90 ms, LBBB widens QRS, LAE and IAB prolong the P wave, and MI lengthens QT. The labels therefore reproduce the reported disease-specific structure at the population level, but this does not establish beat-level label accuracy.

| Class           | $\Delta P$ dur | $\Delta QRS$ | $\Delta PQ$ | $\Delta QT$ | Expected pattern    |
|-----------------|----------------|--------------|-------------|-------------|---------------------|
| <b>AV block</b> | +0.8           | -2.7         | +92.5       | -1.7        | PR prolongation     |
| <b>LBBB</b>     | -0.6           | +10.5        | +9.0        | +4.0        | QRS widening        |
| <b>RBBB</b>     | -6.2           | +2.8         | -8.2        | +10.0       | QRS widening        |
| <b>IAB</b>      | +6.9           | +0.4         | +15.0       | +2.1        | P / PQ prolongation |
| <b>LAE</b>      | +16.6          | -0.3         | +24.0       | +1.8        | P-wave widening     |
| <b>FAM</b>      | +1.6           | -0.6         | +6.3        | +0.2        | mild P change       |

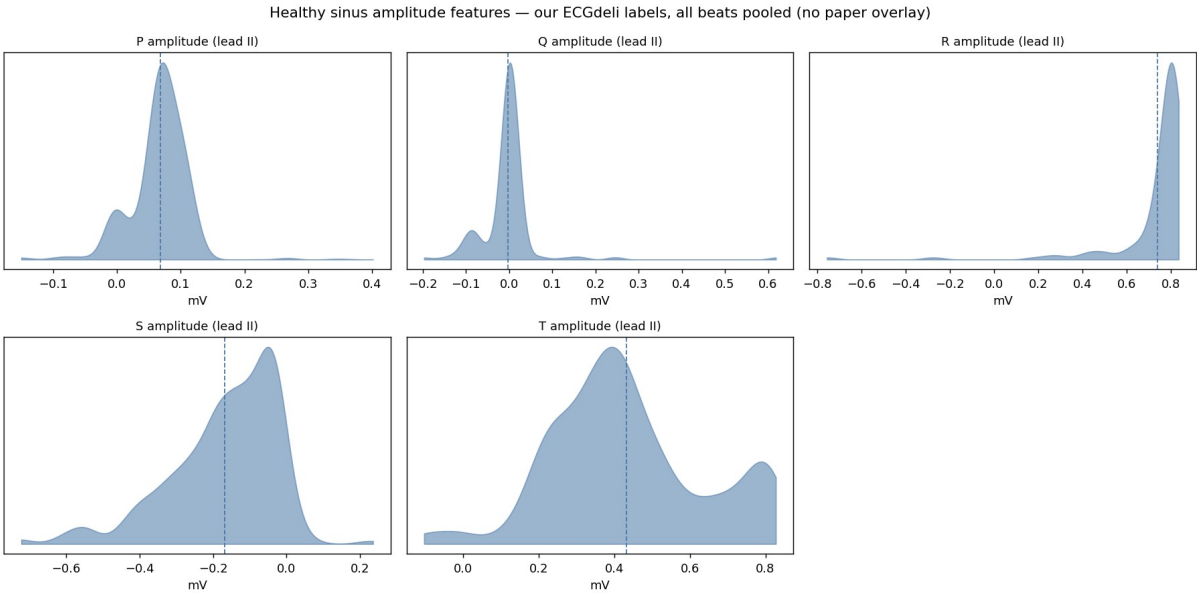


|           |       |      |       |       |                                  |
|-----------|-------|------|-------|-------|----------------------------------|
| <b>MI</b> | +22.5 | +2.4 | +46.8 | +19.2 | QT /<br>repolarisation<br>change |
|-----------|-------|------|-------|-------|----------------------------------|

Per-beat view (no comparison). Before the per-record aggregation, it is worth seeing the raw distributions of our labels with all beats pooled, exactly as first computed. The template-related beats produce narrow, sometimes multi-modal peaks whose shape does not match the paper's per-ECG statistics, so these are shown on their own, without the paper overlay. The per-record comparison follows.



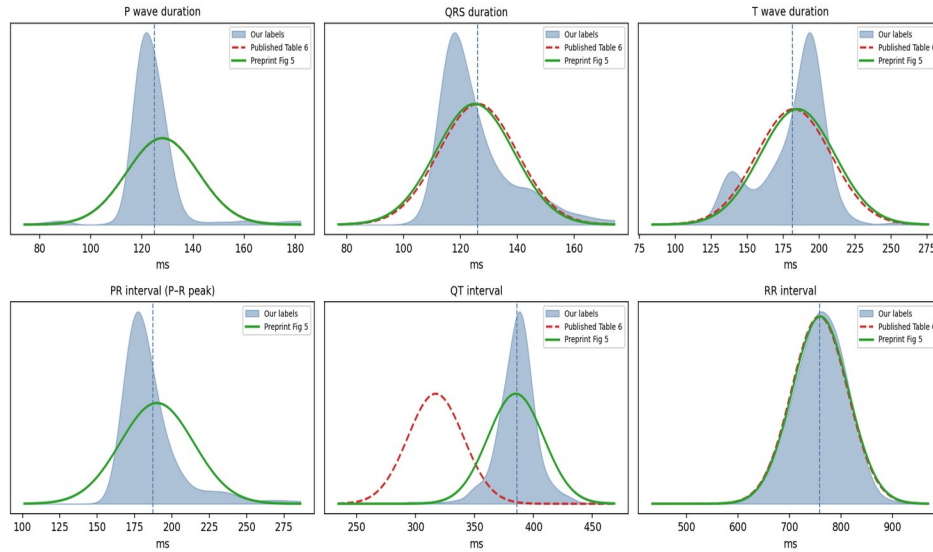
Healthy sinus timing biomarkers (lead II): our labels, all beats pooled, shown without a paper overlay, not directly comparable with the descriptor's per-ECG statistics.



Healthy sinus amplitude features (lead II): our labels, all beats pooled, shown without a paper overlay.

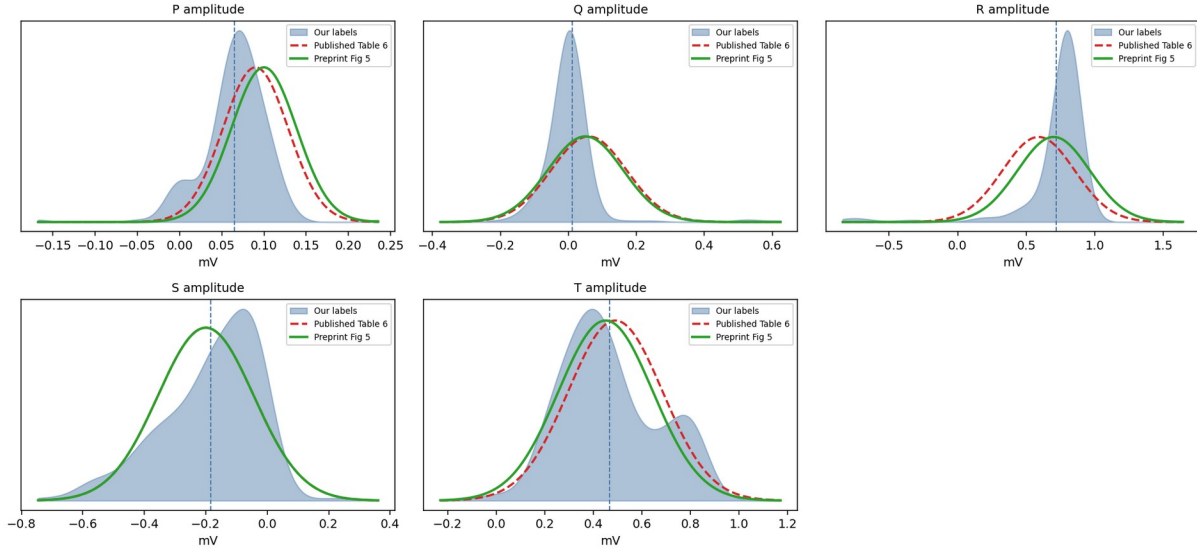
Distributions. Overlaying the density of each healthy biomarker on the reference Gaussians shows broad agreement for several features, not exact equality of shape. Each panel carries the applicable references: the published Table 6 Gaussian (dashed) and the preprint Figure 5 (solid). Because the preprint publishes only probability-density plots (no numeric table), its curves are centred on the Figure 5 values with widths borrowed from the Table 6 standard deviation. Among the directly comparable timing features, P-wave duration, QRS duration, T-wave duration, and RR interval lie close to both reference curves, while QT separates the references. The atrioventricular timing panel instead compares our PRint with the preprint only, because Table 6 reports the different PQint measure; the amplitudes broadly overlap both references.

Healthy sinus, lead II (one median per record): our density vs published Table 6 (dashed) and preprint Figure 5 (solid).  
Of the directly comparable features only QT separates the two references — our density sits on the preprint (~385 ms), not Table 6 (~317 ms). The AV panel shows PRint (P–R peak) vs the preprint; Table 6's PQ interval is a different measure (see §4.4).



*Healthy timing biomarkers (lead II, one value per record): our label densities (shaded) vs the published Table 6 Gaussians (dashed red) and the preprint Figure 5 reference (solid green). Of the directly comparable features (P/QRS/T duration, RR) both references overlap our densities; only QT separates them, with our density on the preprint (~385 ms), not Table 6 (~317 ms). The fourth panel is the preprint's PR interval (P-onset to R-peak, ~190 ms); Table 6 instead reports the PQ interval (P-onset to QRS-onset, ~127 ms) — a different measure shown in the Section 4.4 table — so no Table 6 curve appears on that panel.*

Healthy sinus, lead II amplitudes (one median per record): our density vs published Table 6 (dashed) and preprint Figure 5 (solid). The two references agree on amplitudes; both overlap our densities (P and Q run low, as expected for the simulated beats).



Healthy amplitude features (lead II, one value per record): our densities (shaded) vs the published Table 6 (dashed red) and preprint Figure 5 (solid green) references. The two references agree on the amplitudes, and both overlap our densities.

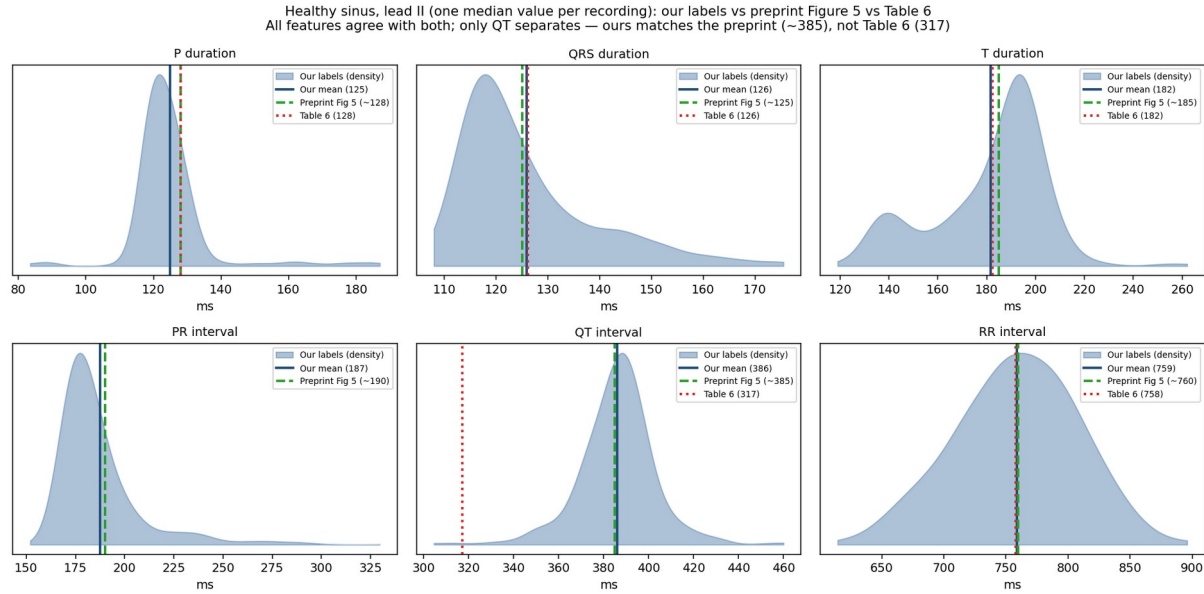
In short, the relabelling reproduces the MedalCare-XL published population statistics across timing means, spreads, amplitudes and disease-class structure, with QT closer to the descriptor's preprint (Section 4.6) than to the published Table 6. Because the paper's features were also produced by ECGdeli, this confirms reproducibility of the pipeline, not absolute accuracy, which is why the QT is compared against the preprint and cross-checked with NeuroKit2 below. Averaging ECGdeli with NeuroKit2 was also tested and moves every biomarker further from Table 6 (the paper's reference is itself ECGdeli-derived), so combining tools was not pursued.

#### 4.6 Comparison against the preprint (per-signal aggregation)

A preprint of the descriptor (arXiv:2211.15997) reports the same feature distributions in its Figure 5. To compare at the population level, we summarise each recording to one value per lead, the median of each interval across its  $\sim 13$  beats, and then take the cohort mean and spread per lead and class. The two levels are, per-signal median (Level 1), then per-lead/per-class summary (Level 2). Each MedalCare recording is a 10 s ECG whose beats share the P/QRS template and vary only in RR and a per-beat stretch of the QRSoff-Toff segment, so a per-recording median matches the paper's per-ECG comparison. The choice of the median specifically is ours (the paper does not state its beat aggregation strategy).

Four directly comparable non-QT timing features—P-wave duration, QRS duration, T-wave duration and RR interval—are close to both published

references. The atrioventricular timing feature differs in definition between the two versions of the descriptor: the preprint reports P<sub>Print</sub>, measured from P-wave onset to the R peak, whereas the published Table 6 reports P<sub>Qint</sub>, measured from P-wave onset to QRS onset. Our corresponding lead-II values are 187 ms for P<sub>Print</sub>, close to the preprint estimate of approximately 190 ms, and 124 ms for P<sub>Qint</sub>, close to the published Table 6 value of 127 ms. QT remains the only directly comparable timing feature showing a large numerical discrepancy between the two references: our value of 386 ms is close to the preprint estimate of approximately 385 ms, but not to the published Table 6 value of 317 ms. Figure 6's per-class shifts also reproduce (wider QRS in RBBB/LBBB, prolonged PR in AV block, wider P in IAB/LAE, prolonged QT in MI), and the R/S/T amplitudes broadly match, while P and Q amplitudes are low in the relevant leads. An independent manual delineation of one ECG by the supervisor gave a lead-II QT of 364 ms, closer to our labels than to 317 ms, but this is one example and is not an independent ground truth.



*Healthy sinus, lead II: per-recording median densities compared with the applicable reference values. P-wave duration, QRS duration, T-wave duration, and RR interval are directly comparable between the preprint and Table 6. The atrioventricular panel compares P<sub>Print</sub> with the preprint only, because Table 6 instead reports P<sub>Qint</sub>. Among the directly comparable features, QT is the only substantial reference discrepancy, with our distribution closer to the preprint.*

### Per-class reproduction (Figure 6).

Beyond the healthy lead-II panels of Figure 5, we also reproduced the per-class contrasts of Figure 6 (disease vs healthy, in the disease-relevant lead). Every timing shift moves in the direction the figure shows: QRS duration widens in RBBB (+5 ms) and LBBB (+12 ms), the PR interval prolongs in AV

block (+94 ms), P duration widens in IAB (+8 ms) and LAE (+11 ms), and QT prolongs in myocardial infarction at V4 (+12 ms). The amplitude panels agree too: MI changes the Q amplitude in lead II (-0.02 to +0.14 mV) and the R amplitude in V2 (-2.41 to -2.90 mV), while fibrotic atrial cardiomyopathy reduces the P-wave amplitude in aVL (0.043 to 0.008 mV) and V6 (0.076 to 0.060 mV); left atrial enlargement leaves the R amplitude unchanged, as expected for an atrial condition. So, the labels reproduce the reported population-level feature changes not only for the healthy population but also for the disease-specific contrasts that Figure 6 was designed to show. (Each Figure 6 amplitude contrast uses a 100-record sample per class; the healthy lead-II amplitude comparison in Section 4.5 uses 200 records, and its all-lead summary is in `amplitudes_alllead_sinus.csv`.)

#### 4.7 Independent second-tool cross-check (NeuroKit2)

As a final, independent check, the ECGdeli labels were cross-checked against NeuroKit2, a separate open-source delineator that uses a discrete wavelet method and shares no code with ECGdeli. The check was run over the entire dataset, all 16,848 records and all twelve leads. Each ECGdeli beat was matched one-to-one to its nearest NeuroKit2 beat by R-peak within tolerance (each NeuroKit2 beat is used at most once); because successive beats are separated by roughly an RR interval, far larger than the matching tolerance, re-running with strict one-to-one matching left every agreement figure unchanged. The number of matched beats varies by fiducial, from about 1.52 million (T-offset) to 2.00 million (R-peak). For each fiducial we recorded the fraction of matched beats on which the two tools agree within a per-fiducial tolerance, together with the median difference (NeuroKit2 minus ECGdeli).

| Fiducial          | Agree %       | Median $\Delta$ (ms) |
|-------------------|---------------|----------------------|
| <b>P onset</b>    | 55.7 %        | +26                  |
| <b>P peak</b>     | 64.9 %        | +4                   |
| <b>P offset</b>   | 60.0 %        | -20                  |
| <b>QRS onset</b>  | 21.1 %        | -44                  |
| <b>Q peak</b>     | 30.6 %        | -24                  |
| <b>R peak</b>     | 46.7 %        | 0                    |
| <b>S peak</b>     | 32.8 %        | +2                   |
| <b>QRS offset</b> | 44.0 %        | -12                  |
| <b>T onset</b>    | 34.6 %        | +56                  |
| <b>T peak</b>     | <b>82.3 %</b> | <b>-2</b>            |
| <b>T offset</b>   | <b>81.5 %</b> | <b>-8</b>            |

Agreement is the share of matched beats within tolerance; a positive median means NeuroKit2 places the point later than ECGdeli. The T-peak and T-offset rows are highlighted. Because R-peaks were used for beat matching, the R-peak row is not an independent validation.

The T-end receives partial independent support. Across the whole dataset and all twelve leads, the two tools agree on the T-peak 82.3 % of the time and on the T-offset 81.5 % of the time within the stated tolerances, with median differences of  $-2$  ms and  $-8$  ms. This shows majority agreement around the T-peak and T-offset, but the medians do not describe the full spread and do not prove that each label is accurate. The result makes a gross systematic T-offset displacement of approximately 70 ms less likely, while the QT difference between Table 6 and the preprint remains unresolved.

Two fiducials show low agreement. NeuroKit2 places the QRS-onset  $\sim 44$  ms earlier than ECGdeli, and the T-onset  $\sim 56$  ms later. Because ECGdeli's QRS onset produces PQ, or clinical PR, intervals close to the published reference, it is retained as the project reference, but neither tool can be treated as ground truth. The T-onset is an ambiguous ST-to-T junction on which delineation conventions may differ. These offsets indicate systematic method differences and may include labelling error.

Both tools may differ when placing P and T boundaries or peaks where waves are small, inverted or biphasic, contributing to the lower agreement in those settings.

**Isolating large inter-tool disagreements.** To focus review beyond the systematic boundary offsets, we kept only beats on which an actual peak (P, R or T) or the T-offset differs by more than 60 ms. This leaves 608,176 beats — about 30 % of matched beats and they cluster where morphology is hardest

| Disease class | Large-disagreement beats | Share  |
|---------------|--------------------------|--------|
| mi            | 392,650                  | 64.6 % |
| rbbb          | 41,274                   | 6.8 %  |
| lae           | 31,725                   | 5.2 %  |
| iab           | 30,519                   | 5.0 %  |
| fam           | 28,774                   | 4.7 %  |
| lbbb          | 28,054                   | 4.6 %  |
| avblock       | 27,618                   | 4.5 %  |
| sinus         | 27,562                   | 4.5 %  |

Nearly two-thirds of these large inter-tool disagreements are myocardial-infarction beats, consistent with MI's altered repolarisation and its biphasic and inverted T waves. By lead they concentrate in aVR (105,940), aVL (81,096), I (68,402) and III (56,641), the low-amplitude and inverted-wave leads, while lead II has the fewest (29,632). The split across P-peak ( $\approx 394$ k), T-offset ( $\approx 261$ k) and T-peak ( $\approx 239$ k) is consistent with small or atypical P and T waves in particular leads and in infarction. These beats will be prioritized for manual review or to down-weight in strict per-lead T-timing evaluation.

## 5.1 Quality-control status carried with each signal

So that signals needing manual correction are trackable directly from the label table, each row carries a qc\_status derived from the per-record-lead analysis of Section 4.3, together with a persistent flag, the list of problem leads, and a review priority

| qc_status | Signals | Meaning / action |
|-----------|---------|------------------|
|-----------|---------|------------------|



|                 |               |                                                                                                                              |
|-----------------|---------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>critical</b> | 2,300 (13.7%) | At least one lead is persistently flagged ( $\geq 50\%$ of its beats flagged). These are prioritized for manual correction.  |
| <b>minor</b>    | 5,512 (32.7%) | Some critical beats but no persistently flagged lead - lower-priority cases that an aligned per-record consensus may absorb. |
| <b>clean</b>    | 9,036 (53.6%) | No critical beats detected by the automated screen; retained subject to later checks.                                        |

The 2,300 critical signals correspond to the 4,251 persistently flagged (record, lead) units (a signal can have several problem leads) and are dominated by myocardial infarction (1,720), then LBBB (131) and RBBB (102). The higher-priority combined review set contains 13,792 record-lead units across 5,601 recordings: it comprises the 4,251 units that meet the persistent rule-based QC threshold and 9,541 additional units that fall below that threshold but contain both rule-based flags and persistent cross-method disagreement, captured in `require_manual_label_priority.csv`.

## Appendix — Code and data availability

All code and configuration for the pipeline described in this report are version-controlled in the project repository (<https://github.com/john-thuo1/delineation->). Run order, file paths and quality-control thresholds are documented in the repository README and in `config/paths.yaml` and `config/pipeline_settings.yaml`; every figure, table and statistic above is reproducible from the scripts listed below. The ECGdeli labelling (the master fiducials table) is the one fixed input, treated as described in Section 3.

Delineation driver (Section 3): `ecgdeli_labelling/scripts/run_ecgdeli_medalcare.m`

Quality control (Sections 4.1–4.2): `ecgdeli_labelling/scripts/qc_review_list.py`

ECGdeli preprocessing sweep and scoring (Section 4.4):  
`ecgdeli_labelling/scripts/ecgdeli_config_sweep.m` and `statistics/scripts/score_vs_table6.py`

Population statistics and figures (Sections 4.5–4.6):  
`statistics/scripts/build_per_signal_stats.py`, `build_comparison_figure.py`,  
`build_gaussian_overlays.py` and `reproduce_paper_stats.py`

Independent cross-checks (Section 4.7): `ecgdeli_labelling/scripts/crosscheck_neurokit.py`,  
`crosscheck_neurokit_peak.py` and `build_review_targets.py`

Training-set assembly and QC columns (Section 5):  
`dataset_curation/scripts/build_master.py`, `add_qc_status.py`, `add_crosscheck_qc.py` and  
`build_manual_worklist.py`

Single-record signal-and-label views (which regenerate on demand the kind of per-signal example shown during development): `dataset_curation/scripts/extract_signal.py`

Data. The raw 12-lead signals are the MedalCare-XL release; the ECGdeli toolbox is from KIT-IBT ([github.com/KIT-IBT/ECGdeli](https://github.com/KIT-IBT/ECGdeli)). The large derived tables (the master fiducials,

master\_labels.csv and the NeuroKit disagreement files) are regenerable with the scripts above and are therefore excluded from the repository, while the small report-featured summary tables are included.